Apparent Late-Time Acceleration Without a Cosmological Constant

Closure-Stratified Late-Time Cosmology in Darke Theory

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Abstract

Persistent late-time acceleration is usually compressed into a cosmological-constant or dark-energy sector. The present paper develops a harder Darke-theoretic alternative. The central claim is not merely that a primitive cosmological constant is unnecessary, but that the current Darke Theory corpus already favors a completed no-Λ late-time cosmology. The reason is twofold. First, the core cosmological construction already writes late-time expansion as the sum of a fading recursion contribution and an informational-drift contribution rather than as a vacuum-energy law. Secondly, the current post-threshold bridge construction does not naturally provide a clean interior post-threshold vacuum that could act as a true effective-Λ floor.

A completed minimal late-time model is then stated. The late universe is partitioned into low-closure void-like domains and high-closure bound domains. A canonical shell-averaged Hubble law is defined by

$$\bar{H}(z) = H_B^\infty + R_0(1+z)^m + \frac{\Delta H_\infty}{1 + K_0(1+z)^p},$$

where the first term is the bound-sector drift baseline, the second term is the fading recursion contribution, and the third term is the closure-stratified volume-dominance contribution. The corresponding deceleration law is derived explicitly and shown to admit apparent late-time acceleration without any cosmological constant. The model is then completed observationally by allowing the stratification scale parameter to depend on void structure, closure-legibility structure, and a preferred directional mode. This yields an exact residual deformation law of the form

$$R_x(\hat{n}, z) = A(z)V(\hat{n}) + B(z)\Xi(\hat{n}) + C(z)\,\hat{n}\cdot d + \epsilon(\hat{n}, z),$$

thereby realizing the structural-commensurability framework of the wider corpus directly from the no-Λ late-time cosmology itself.

The result is a theorem-first completed model, not yet a data-fitting paper. Its exact claim is that within the current Darke Theory corpus, apparent late-time acceleration can be generated by closure-stratified volume dominance on top of a fading recursion sector and a drift background, while any fitted Λ_{app} is demoted from ontological constant to scalar-fit compression residue.

1 Introduction

Standard late-time cosmology is often compressed into one scalar story: once ordinary matter and radiation dilute, an additional cosmological-constant or dark-energy sector dominates and

drives apparent acceleration. Within Darke Theory, that explanatory order is not the natural one. The relevant sequence is already fixed elsewhere in the corpus:

$$\text{recursive differentiation} \Longrightarrow \text{closure} \Longrightarrow \text{admissibility} \Longrightarrow \text{persistent structure.}$$

Stable cosmological inference is therefore not granted by default. It must be earned by structurally commensurate observational reduction and by a cosmological law that fits the load-bearing objects of the theory itself.

The present paper takes the strongest current route. It does not attempt to soften the anti-Λ posture by introducing a merely nicer vacuum term. Instead it closes the late-time object in the opposite direction. The core cosmological construction already writes expansion as the sum of a recursion term and an informational-drift term. The ontological foundation already states that low-closure regions expand preferentially and that apparent acceleration emerges once coherence stratification becomes significant. The structural-commensurability paper already shows that scalar expansion inference is valid only when the conditioned residual field is structurally null. The present paper combines those strands into one completed minimal late-time cosmology.

The paper has six purposes:

(i) to record why the current corpus does not naturally support a clean effective-Λ vacuum branch,

(ii) to state a completed canonical no-Λ late-time Hubble law,

(iii) to derive the corresponding effective deceleration law,

(iv) to complete the directional and environment-conditioned deformation sector,

(v) to derive the distance-modulus residual law from that completed late-time cosmology,

(vi) to state precisely what a fitted Λ_{app} means in this framework.

This paper is theorem-first and model-definitional in scope. It does not yet calibrate the late-time parameters against survey likelihoods. What it does claim is exact: the current Darke Theory corpus supports a completed no-Λ late-time cosmology in which apparent acceleration is generated by closure-stratified volume dominance on top of a fading recursion term and a drift baseline, while fitted scalar dark-energy sectors are demoted to compression residue.

2 Foundational setting and claim boundary

Only the minimum load-bearing commitments of the wider corpus are required here.

Definition 1 (Late-time split)**.** The canonical late-time Darke-theoretic expansion split is

$$H(z) = H_{\text{rec}}(z) + H_{\text{flow}}(z),$$

where H_{rec} is the recursion-side contribution and H_{flow} is the informational-drift contribution.

The broader cosmological paper is explicit that neither a cosmological constant nor a primitive dark-energy sector is introduced there. Late-time behavior is instead associated with a fading

recursion contribution and increasing dominance of informational wave drift. This already removes any structural necessity for primitive Λ-ontology.

A second strand of the corpus sharpens the same conclusion. The ontological foundation states that low-closure regions expand preferentially and that once coherence stratification becomes significant, apparent acceleration appears as low-closure regions come to dominate the volume fraction. A third strand then fixes the observational discipline: scalar expansion reduction is valid only under structural commensurability, that is, only when the conditioned residual field is structurally null. If directional or environment-conditioned residual structure survives, one scalar late-time expansion parameter is not a regime invariant by default.

The present paper therefore adopts the following exact claim boundary.

Definition 2 (No-Λ late-time branch)**.** A late-time cosmological branch is called *no-Λ* if its expansion law contains no primitive cosmological-constant sector and no clean emergent effective vacuum floor, while still admitting apparent acceleration through closure-stratified dynamics and observational compression.

3 Why the current bridge does not provide a clean effective-Λ vacuum

Before stating the completed late-time model, it is necessary to record why the present paper does not attempt to salvage a true effective-Λ vacuum from the current post-threshold bridge.

In the current bridge construction, the geometry-activation coupling is

$$F(\Phi) = \sigma(\chi)(1 + \xi\chi + \eta\chi^2), \qquad \chi = \frac{\Phi - \Phi_{\text{crit}}}{\Phi_0},$$

with $F(\Phi) > 0$ throughout the post-threshold regime $\Phi > \Phi_{\text{crit}}$. The coherence potential is written explicitly as

$$V(\Phi) = \Lambda_0 + \frac{\mu^2}{2}(\Phi - \Phi_{\text{crit}})^2 + \frac{\lambda}{4}(\Phi - \Phi_{\text{crit}})^4.$$

The weak-field and perturbative-vacuum constructions are stated around a stationary vacuum satisfying

$$V'(\Phi_*) = 0, \qquad V''(\Phi_*) > 0, \qquad \Phi_* > \Phi_{\text{crit}}.$$

However,

$$V'(\Phi) = (\Phi - \Phi_{\text{crit}})\Big(\mu^2 + \lambda(\Phi - \Phi_{\text{crit}})^2\Big),$$

and under the natural stability reading $\mu^2 > 0$ and $\lambda > 0$, the only stationary point lies at

$$\Phi = \Phi_{\text{crit}},$$

that is, on the threshold boundary rather than in the interior post-threshold regime.

Proposition 1 (No clean interior post-threshold vacuum in the current bridge)**.** *Under the natural stable-sign reading $\mu^2 > 0$ and $\lambda > 0$ of the current bridge potential, the present bridge does not supply a clean interior post-threshold stationary vacuum that could act as a true effective-Λ floor.*

Proof. The derivative $V'(\Phi)$ vanishes only when $\Phi = \Phi_{\text{crit}}$. But the bridge's geometric and

relativistic claims apply only on the open post-threshold domain $\Phi > \Phi_{\text{crit}}$. Hence the only natural stationary point lies on the regime boundary rather than within the admissible interior post-threshold domain. Therefore the current bridge does not provide a clean interior post-threshold vacuum floor. □

This is why the present paper takes the harder route. The finished late-time cosmology will not be built by introducing a softened vacuum floor. It will be built directly as a no-Λ closure-stratified late-time law.

4 Canonical completed late-time model

4.1 Two effective domain classes

Work only on the late-time observational regime and partition the universe into two coarse domain classes:

- V: low-closure, void-like domains,
- B: high-closure, bound or closure-heavy domains.

Let the effective closure scores of these classes be frozen over the late-time observational window:

$$0 \leq c_V < c_B \leq 1.$$

Let the shared late-time drift background be a smooth function $H_d(z)$ and define the local closure-to-expansion map

$$S(c) := \upsilon(1-c) - \omega c^{\nu}, \qquad \upsilon > 0, \quad \omega > 0, \quad \nu \geq 1. \tag{4.1}$$

Then the domain expansion rates are

$$H_V(z) = H_d(z) + S(c_V), \qquad H_B(z) = H_d(z) + S(c_B). \tag{4.2}$$

Since

$$S'(c) = -\upsilon - \omega\nu c^{\nu-1} < 0,$$

we obtain immediately:

$$H_V(z) - H_B(z) = S(c_V) - S(c_B) > 0. \tag{4.3}$$

Thus low-closure domains always expand faster than high-closure domains.

4.2 Completed late-time closure-stratification law

To close the model completely, define the low-closure volume fraction by

$$\bar{f}(z) = \frac{1}{1 + K_0(1+z)^p}, \qquad K_0 > 0, \quad p > 0. \tag{4.4}$$

As z decreases, $(1+z)^p$ decreases, so $\bar{f}(z)$ rises. This is the completed constitutive implementation of the ontological claim that low-closure regions come to dominate the late-time volume fraction.

Now freeze the shared drift background itself in the minimal canonical form

$$H_d(z) = H_B^\infty + R_0(1+z)^m, \qquad H_B^\infty > 0, \quad R_0 > 0, \quad m > 0. \tag{4.5}$$

Here H_B^∞ is the asymptotic bound-sector drift baseline and $R_0(1+z)^m$ is the fading recursion contribution. The canonical shell-averaged completed late-time Hubble law is then

$$\boxed{\bar{H}(z) = H_B^\infty + R_0(1+z)^m + \frac{\Delta H_\infty}{1 + K_0(1+z)^p}} \tag{4.6}$$

with

$$\Delta H_\infty := S(c_V) - S(c_B) > 0. \tag{4.7}$$

Equation (4.6) is the canonical completed isotropic no-Λ late-time law.

5 Acceleration without a cosmological constant

Differentiate (4.6):

$$\bar{H}'(z) = mR_0(1+z)^{m-1} - \frac{pK_0\Delta H_\infty(1+z)^{p-1}}{\left(1 + K_0(1+z)^p\right)^2}. \tag{5.1}$$

Hence the effective deceleration parameter is

$$\boxed{q(z) = -1 + \frac{(1+z)\bar{H}'(z)}{\bar{H}(z)}} \tag{5.2}$$

that is,

$$q(z) = -1 + \frac{mR_0(1+z)^m - \dfrac{pK_0\Delta H_\infty(1+z)^p}{\left(1 + K_0(1+z)^p\right)^2}}{\bar{H}(z)}. \tag{5.3}$$

The second contribution in the numerator is strictly negative. It is the closure-stratification acceleration driver. Apparent acceleration occurs whenever this negative contribution outweighs the fading recursion slope strongly enough to push $q(z)$ below zero.

Theorem 1 (Completed no-Λ late-time acceleration theorem)**.** *In the canonical completed late-time model (4.6), apparent late-time acceleration can occur without any cosmological constant. It is generated by the closure-stratification term*

$$-\frac{pK_0\Delta H_\infty(1+z)^p}{\left(1 + K_0(1+z)^p\right)^2}$$

in the effective slope of $\bar{H}(z)$.

Proof. Equation (5.3) gives the deceleration parameter explicitly. No cosmological-constant term appears in either $\bar{H}(z)$ or its derivative. The only negative slope contribution in the completed model comes from the growth of the low-closure volume fraction encoded in the

closure-stratification term. Therefore if $q(z) < 0$, the apparent acceleration is generated by closure stratification rather than by a cosmological constant. □

6 Directional and environment-conditioned completion

The isotropic background is not the full story. The wider corpus already fixes the minimal late-time residual deformation architecture in terms of a void field, a closure-legibility field, and a preferred directional mode. We therefore complete the late-time cosmology by deforming the stratification scale parameter:

$$\boxed{K(\hat{n}) = K_0\Big[1 - \alpha V(\hat{n}) - \beta\Xi(\hat{n}) - \gamma\,\hat{n}\cdot d\Big]} \tag{6.1}$$

with $K(\hat{n}) > 0$.

The full directional late-time Hubble field is then

$$\boxed{H(\hat{n}, z) = H_B^\infty + R_0(1+z)^m + \frac{\Delta H_\infty}{1 + K(\hat{n})(1+z)^p}.} \tag{6.2}$$

Linearizing around K_0 yields the first-order deformation

$$\delta H(\hat{n}, z) = \frac{\Delta H_\infty K_0 (1+z)^p}{\big(1 + K_0(1+z)^p\big)^2}\Big[\alpha V(\hat{n}) + \beta\Xi(\hat{n}) + \gamma\,\hat{n}\cdot d\Big]. \tag{6.3}$$

This is the finished late-time deformation field.

7 Distance-modulus residual law

Let the isotropic reference luminosity distance induced by the completed background law be

$$D_*(z) = (1+z)\int_0^z \frac{dz'}{\bar{H}(z')}. \tag{7.1}$$

To first order in the Hubble deformation, the distance-modulus residual field is

$$R_x(\hat{n}, z) \approx -\frac{5}{\ln 10}\,\frac{1+z}{D_*(z)}\int_0^z \frac{\delta H(\hat{n}, z')}{\bar{H}(z')^2}\,dz' + \epsilon(\hat{n}, z). \tag{7.2}$$

Substituting (6.3) gives

$$\boxed{R_x(\hat{n}, z) = A(z)V(\hat{n}) + B(z)\Xi(\hat{n}) + C(z)\,\hat{n}\cdot d + \epsilon(\hat{n}, z),} \tag{7.3}$$

where

$$A(z) = -\frac{5\alpha}{\ln 10}\frac{1+z}{D_*(z)}\int_0^z \frac{\Delta H_\infty K_0(1+z')^p}{\left(1+K_0(1+z')^p\right)^2 \bar{H}(z')^2}\,dz',$$

$$B(z) = -\frac{5\beta}{\ln 10}\frac{1+z}{D_*(z)}\int_0^z \frac{\Delta H_\infty K_0(1+z')^p}{\left(1+K_0(1+z')^p\right)^2 \bar{H}(z')^2}\,dz',$$

$$C(z) = -\frac{5\gamma}{\ln 10}\frac{1+z}{D_*(z)}\int_0^z \frac{\Delta H_\infty K_0(1+z')^p}{\left(1+K_0(1+z')^p\right)^2 \bar{H}(z')^2}\,dz'.$$

Thus the structural-commensurability residual architecture of the wider corpus is realized directly by the completed no-Λ late-time cosmology.

Theorem 2 (Completed residual deformation theorem)**.** *The canonical completed no-Λ late-time cosmology implies a distance-modulus residual field of the exact late-time structural form*

$$R_x(\hat{n}, z) = A(z)V(\hat{n}) + B(z)\Xi(\hat{n}) + C(z)\,\hat{n}\cdot d + \epsilon(\hat{n}, z).$$

Therefore the structural-commensurability framework is not merely compatible with the late-time no-Λ cosmology but is generated by it directly.

Proof. The result follows by substituting the completed Hubble deformation law (6.3) into the first-order distance-modulus residual formula (7.2) and grouping the three independent directional structures. The coefficients $A(z)$, $B(z)$, and $C(z)$ are exactly the corresponding line-of-sight kernels. □

8 Meaning of a fitted Λ_{app}

The point of the present construction is not only that acceleration can occur without a cosmological constant. It is also that if an observer insists on compressing the completed late-time universe into an isotropic FLRW+Λ model, the returned scalar dark-energy sector is not an ontological constant.

Definition 3 (Apparent cosmological constant)**.** Let Λ_{app} denote the scalar parameter returned by an imposed isotropic FLRW+Λ fit to a universe actually governed by the completed no-Λ late-time cosmology.

Then the proper interpretation is:

$$\boxed{\Lambda_{\text{app}} = \Lambda_{\text{strat}} + \Lambda_{\text{compress}},} \tag{8.1}$$

where

- Λ_{strat} is the scalar-fit image of the closure-stratified late-time term already present in (4.6) and (5.3),
- $\Lambda_{\text{compress}}$ is the additional scalar-fit compression residue produced if the residual field (7.3) is not structurally null.

Corollary 1 (Apparent Λ as compression residue)**.** *In the completed no-Λ late-time cosmology, a fitted positive Λ_{app} is not a vacuum constant. It is the scalar-fit image of closure-stratified late-time expansion and, where structural commensurability fails, additional compression residue of the residual deformation field.*

Proof. The completed late-time Hubble law (4.6) contains no cosmological constant. The apparent acceleration of the background law is generated by the closure-stratification term in (5.3). The residual field (7.3) is structurally non-null whenever any of V, Ξ, or $\hat{n} \cdot d$ remains active after admissible conditioning. Therefore any scalar FLRW+Λ fit compresses both the background closure-stratified acceleration and, if present, the non-null residual deformation into an apparent scalar dark-energy sector. That fitted quantity is therefore compression residue rather than vacuum ontology. □

9 Falsifiability and exact claim boundary

The present paper closes the structural object, but it does not yet perform parameter estimation. The exact claim boundary is therefore as follows.

What has been established

(1) The current Darke Theory corpus naturally favors a no-Λ late-time cosmology.

(2) The completed canonical late-time Hubble law is (4.6).

(3) The corresponding deceleration law is (5.3), and apparent acceleration can occur without a cosmological constant.

(4) The full directional and environment-conditioned late-time deformation field is (6.3).

(5) The distance-modulus residual law is (7.3), matching the structural-commensurability framework directly.

(6) A fitted Λ_{app} is properly interpreted as compression residue, not ontological vacuum energy.

What is not yet claimed

(1) No survey-level calibration of the parameters

$$H_B^\infty,\ R_0,\ m,\ \Delta H_\infty,\ K_0,\ p,\ \alpha,\ \beta,\ \gamma$$

has yet been performed here.

(2) No claim is made that the present completed minimal law is the unique possible late-time no-Λ realization of Darke Theory. The claim is that it is the canonical completed minimal one supported by the current corpus.

(3) No claim is made that all observed expansion disagreement is already numerically resolved by this model. The present claim is structural and theorem-first.

What would count against the model

The completed late-time model would be weakened or falsified if any of the following occurred:

(a) the residual field becomes structurally null after admissible conditioning while a robust, domain-invariant scalar dark-energy sector remains necessary;

(b) low-closure or void-conditioned sectors fail to correlate with expansion residual structure;

(c) the low-closure volume fraction fails to increase across the late-time observational window;

(d) apparent acceleration persists in a regime where the completed stratification term in (5.3) is forced negligibly small by data.

10 Conclusion

This paper has completed the minimal no-Λ late-time Darke-theoretic cosmology.

The result is exact. The current corpus does not naturally support a primitive cosmological constant, and it does not naturally support a clean interior post-threshold effective-Λ vacuum either. The finished late-time object instead consists of three pieces: a bound-sector drift baseline, a fading recursion contribution, and a closure-stratified volume-dominance term. Their completed combination is

$$\bar{H}(z) = H_B^\infty + R_0(1+z)^m + \frac{\Delta H_\infty}{1 + K_0(1+z)^p}.$$

From this one obtains the exact deceleration law, the exact directional and environment-conditioned Hubble deformation field, and the exact residual deformation law in the same structural form already fixed elsewhere in the corpus.

The overall conclusion is therefore stronger than the mere denial of dark energy. Apparent late-time acceleration is not treated here as evidence for a vacuum constant. It is treated as the scalar-fit image of a universe whose late-time expansion is closure-stratified, void-conditioned, and observationally non-commensurate by default. In that setting, a fitted Λ_{app} is not a deep ontological ingredient. It is the compression residue of the completed late-time structure.

References

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